# Daniel Hardesty Lewis

*Staff ML Engineer – production training, inference, evaluation, reliability*

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

## Core Expertise

8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, with evaluation harnesses that catch regressions and improve production reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

## Technologies

| | |
|---|---|
| Modeling | PyTorch; TensorFlow; scikit-learn; Hugging Face Transformers; LoRA/QLoRA/PEFT |
| LLM apps | embeddings; RAG; LangChain; LangGraph; DSPy; FAISS |
| Data stack | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase |
| Infra | Linux/Unix; Bash; Git; Docker; Kubernetes; Ray; Spark; Dask; Airflow; CI/CD; observability |
| Additional | CUDA; C++; TypeScript; React |

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Properlytic*, New York, NY

Building a consumer real-estate pricing product focused on individual home price forecasting.

- Deployed a current-price prediction model for Manhattan condos and co-ops; achieved 12% MAPE on a holdout set and matched Zillow at 8.4% MAPE under an internal apples-to-apples evaluation protocol.
- Architected a VAE-based latent representation layer over property, transaction, and neighborhood signals to stabilize learning and improve generalization under sparse and noisy features.
- Prototyped a diffusion-based forecasting backbone in Houston; measured 8% annualized compounding error even through regime shifts such as COVID.
- Migrated market-specific models onto the diffusion backbone for standardized training and evaluation.
- Refined product positioning and execution milestones via strategic feedback from a16z, MetaProp, and others.
- Pitch deck: link

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Engineered a what-if framework to analyze BOM cost impacts, lead times, and compliance risks.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.
- Executing customer discovery with global importers/exporters to iterate on distribution and compliance pricing models.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a 0.5 m statewide seamless Texas DEM mosaic using nearest-neighbor stitching across 70+ state, federal, and local elevation sources.
- Developed the web distribution platform end-to-end (data pipeline, tiling, hosting, delivery UX) to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

*155 Claremont Ave – New York, NY 10027 – United States*

*+1 (713) 371-7875* • *daniel@summitgeospatial.com* • *dlewis.ai* • *dhardestylewis*

*dhardestylewis* • *Google Scholar*

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Co-founded and launched summitgeospatial.com to commercialize state-level geomorphology IP, providing precision decision-support for civil engineering and disaster planning.
- Led the source-to-frontend data pipeline for the $40M Texas Disaster Information System (TDIS), integrating large-scale geospatial assets for state-level resiliency.
- Led inter-agency RFP and MOU development with state and federal partners incl. USACE, Texas GLO, translating technical requirements into scope and milestones enabling the $40M resiliency platform.
- Scaled climate and flood models on supercomputers, executing million-node distributed jobs while managing multi-million-dollar compute budgets and federal partnerships (NOAA, NSF, USCE, GLO).
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Developed efficient methods to produce high-resolution (1m) flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.

### Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Targetting theory publications to DeepSeek's GRPO algorithm

Jan. 2025 – Present **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY

- Developed a multi-asset conditional variational autoencoder (CVAE) latent factor model with Skew-T Mixture priors; achieved 85% $R^2$ vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

### Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization.

## Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

## Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

## Key Projects

DeepSeek Math benchmarking and finetuning of Deepseek's GRPO algorithm

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

## Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

## Languages

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Spanish Native Proficiency French Professional Working Proficiency

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